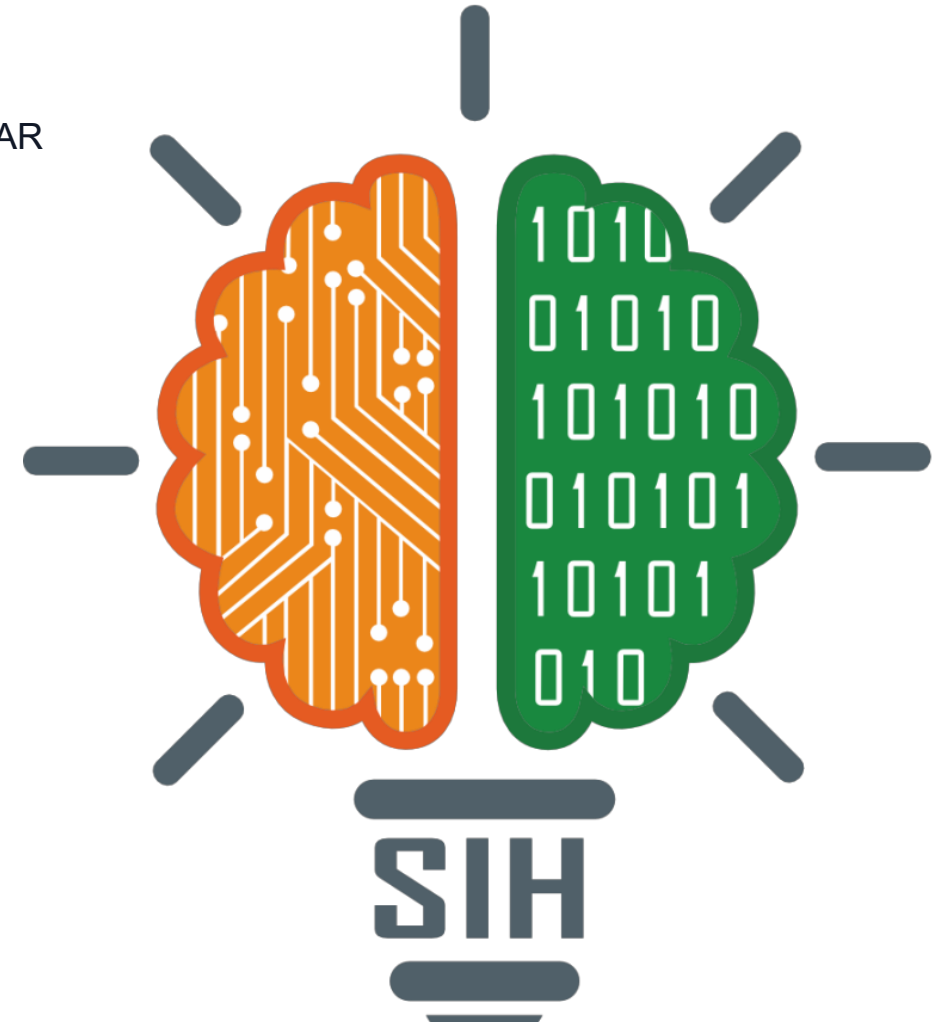


Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation

- Problem Statement ID – 26053
- Problem Statement Title – Adaptive Foveated 2.5D Semantic Mapping from LiDAR Point Clouds for Autonomous Navigation
- Theme – Smart Vehicles
- PS Category – Software
- Team ID – TBA
- Team Name – TeraPulse
- Team Members – Abhay Verma (Team Leader), Lokendra Singh, Yashvardhan Jain, Chetan Meena, Sachin Chaubey, Esha Verma
- Institute – Indian Institute of Information Technology Bhopal





Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation



❖ Proposed Solution

2.5D semantic elevation map generated from raw LiDAR point clouds.

High resolution near the vehicle (0.25 m) for critical near-field perception.

Lower resolution for distant regions (0.75 m & 2.0 m) — cuts unnecessary computation.

Preserves height, terrain gradient & obstacle boundaries without full 3D voxels.

Identifies drivable ground, static obstacles & dynamic objects in real time.

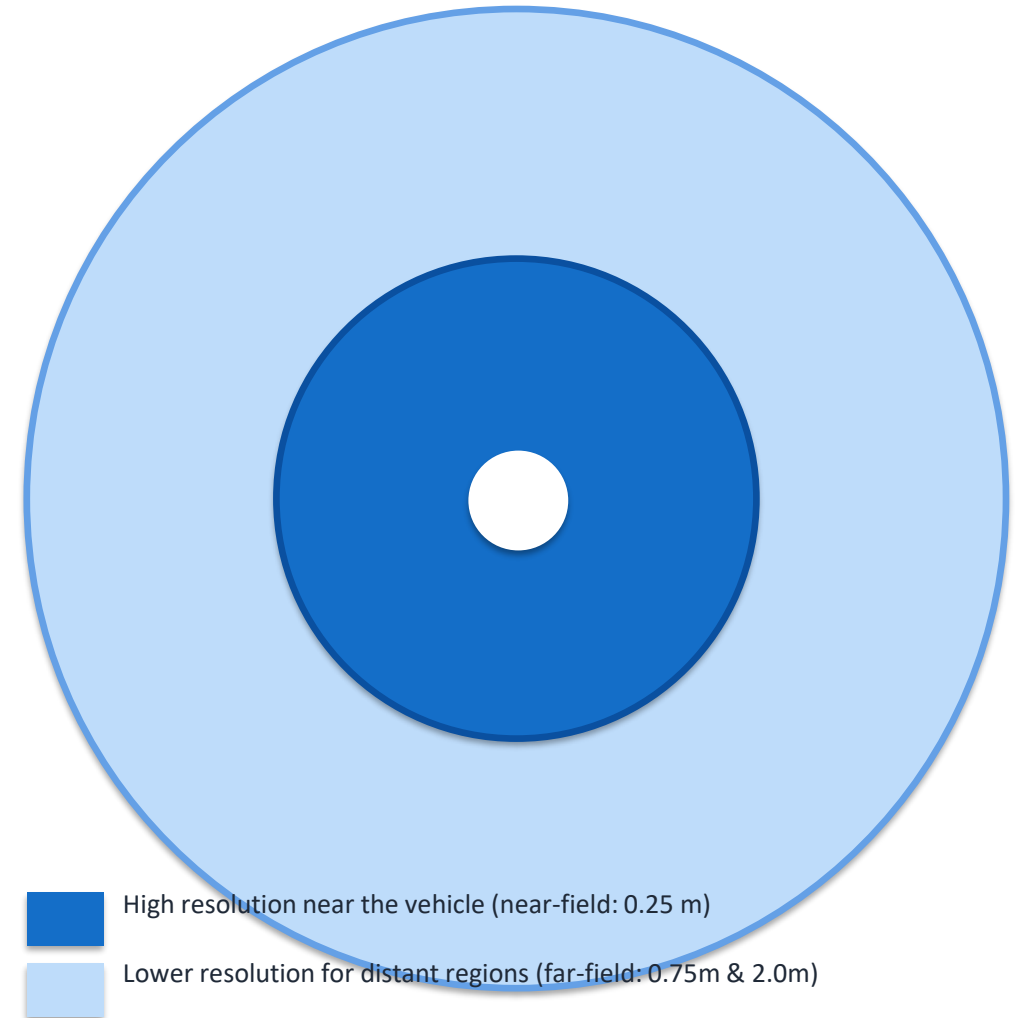
Enables fast closed-loop environmental understanding for path planning.

❖ Innovation & Uniqueness

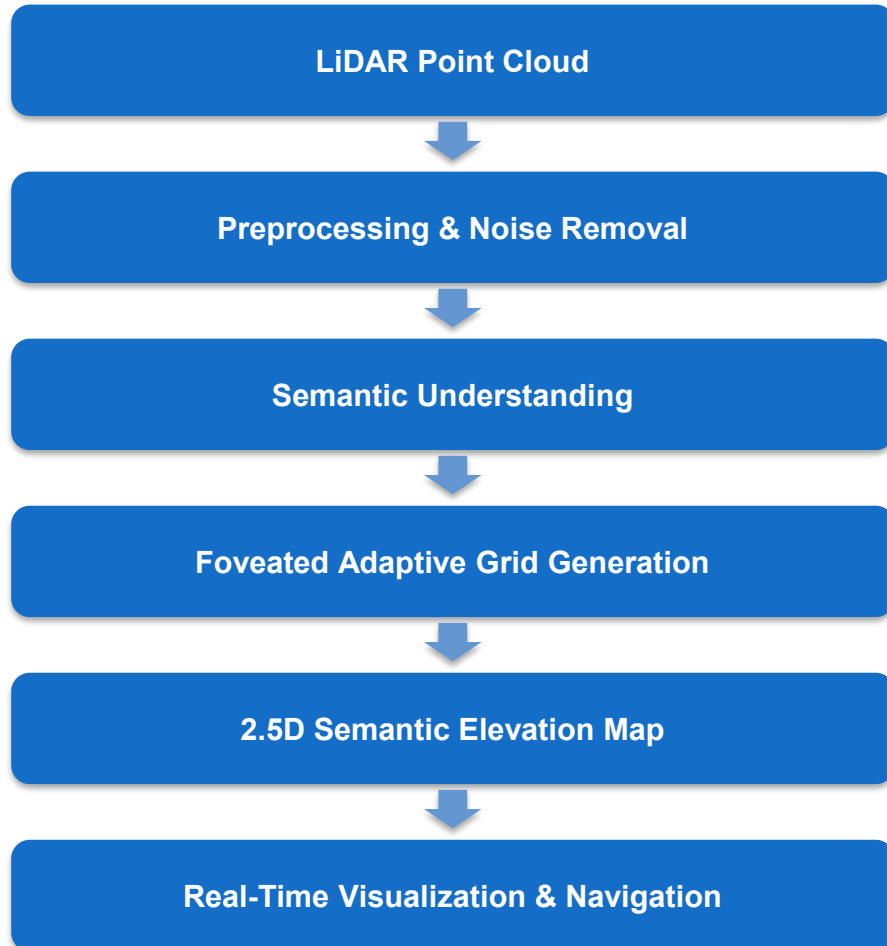
Foveated adaptive resolution focuses computational resources where they matter most, instead of processing the entire environment at the same resolution.

- Slashing cell memory by 66.2% and processing time to 24.23 ms (>35 FPS).

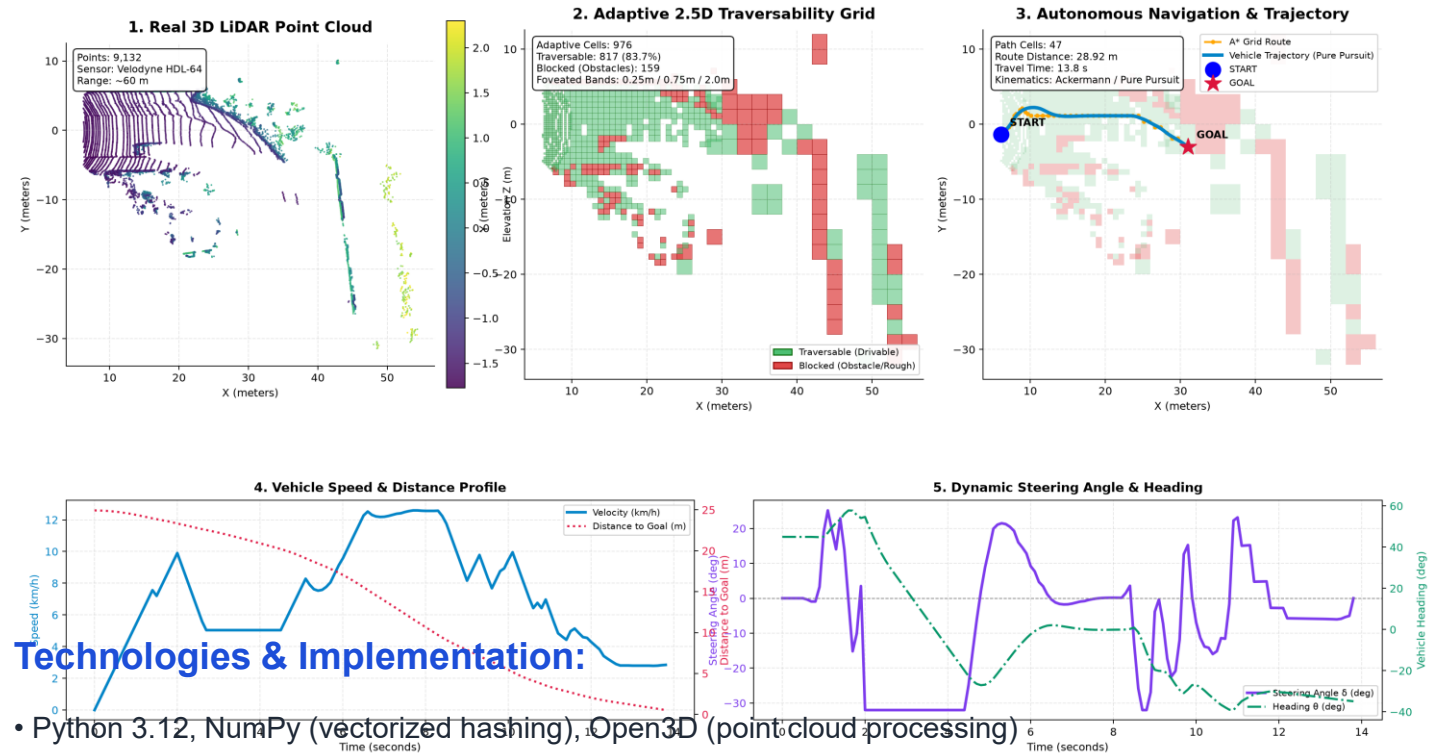
Foveated 2.5D Semantic Elevation Map



System Workflow



TERA PULSE — SIH26053 Autonomous Navigation System Adaptive Variable-Resolution 2.5D LiDAR Mapping & Dynamic Control



Technologies & Implementation:

- Python 3.12, NumPy (vectorized hashing), Open3D (point-cloud processing)
- Deep Learning-Based Semantic Segmentation (Ground, Obstacles, Dynamic Vehicles)
- A* Graph Search & Pure Pursuit Kinematics (Catmull-Rom splines, Ackermann model)
- Real-Time Web Dashboard (Flask REST API, 60 FPS Canvas with moving foveation bubble)

Feasibility

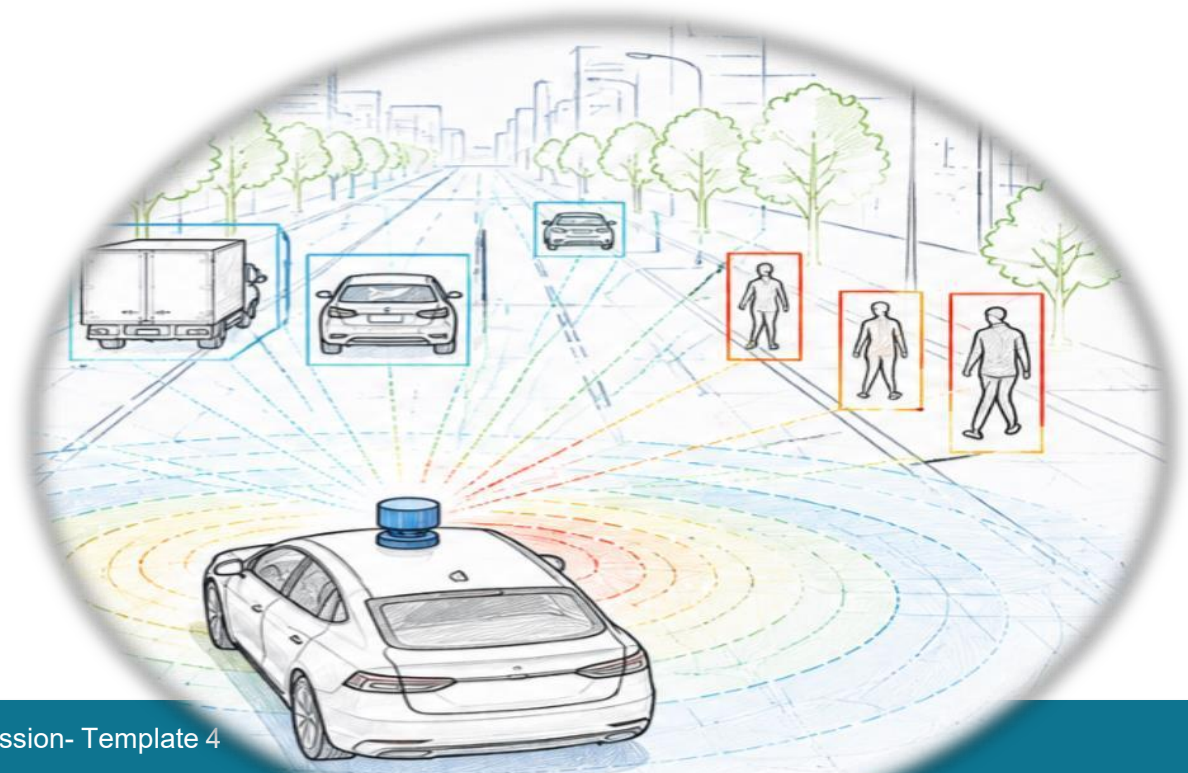
- Uses widely available LiDAR point cloud datasets and open-source tools
- Python, Open3D, and PyTorch enable rapid prototype development
- Adaptive resolution reduces memory and computational requirements
- Modular pipeline allows components to be developed and tested independently

Challenges

- Real-time processing of large LiDAR point clouds at 10–20 Hz
- Accurate semantic classification in unstructured off-road scenes
- Avoiding false-positive obstacles for underpasses & overhead foliage
- Maintaining critical safety detail while reducing distant cell resolution

Mitigation Strategy

- Foveated variable-resolution mapping concentrates power in near-field
- Efficient preprocessing (SOR & RANSAC ground fit) filters dust & noise
- Vertical beam clearance filter enables underpass & tunnel passage
- Validated on 20,672 points in 24.23 ms with 5/5 passing unit tests



Impact

- Supports safer, faster, and reliable autonomous UGV navigation
- Provides sub-meter precision in safety-critical near-field zones
- Active LiDAR penetrates smoke, fog & night darkness for defence
- Enables true real-time terrain and obstacle understanding

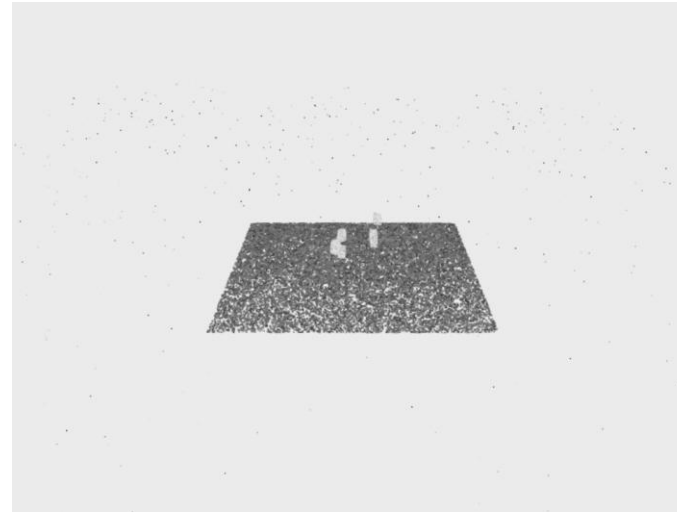
Applications

- Defence UGVs (border surveillance, minefield reconnaissance, logistics)
- Off-road autonomous vehicles, mobile robots & SAR rovers
- Intelligent driver assistance (ADAS HUD in dense Himalayan fog)

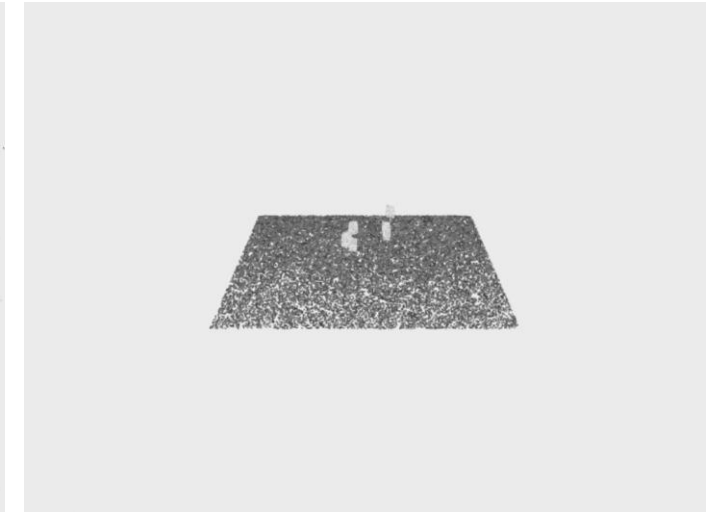
Benefits

- 66.2% lower memory footprint & 4.8× faster latency (24.23 ms)
- Preserves critical 2.5D height profiles (slope, roughness, Z_max)
- Runs smoothly on low-power edge hardware (NVIDIA Jetson Orin)

Visual Impact: Raw Input vs Cleaned Adaptive Perception



Raw LiDAR (20,672 points)



Filtered 2.5D Traversability Map



Production Performance Benchmark on KITTI Velodyne Data:

- Active Stored Cells: Reduced from 2,888 (uniform) to 976 (adaptive) — 66.2% Cell Reduction
- Mapping Compute Time: Reduced from 115.94 ms to 24.23 ms — 4.8× Latency Speedup
- Traversable Area: 817 safe cells (83.71%) accurately identified without collision risk
- Autonomous Execution: Zero-collision 47-cell A* path tracked smoothly via Pure Pursuit

Research Areas

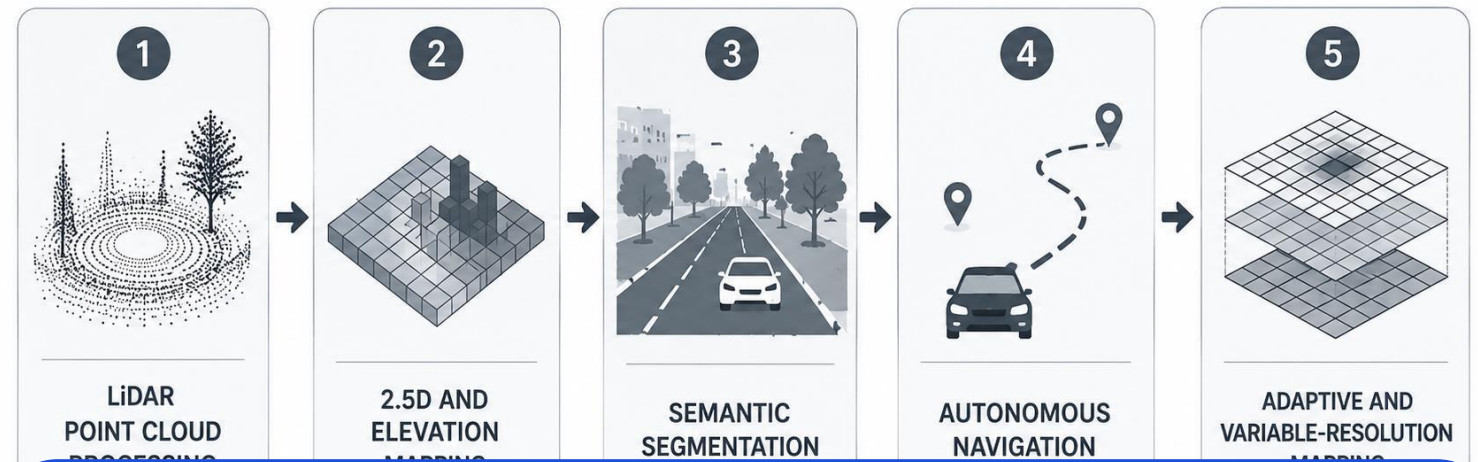
- LiDAR point cloud processing & filtering
- 2.5D and elevation mapping for rough terrain
- Deep semantic segmentation of point clouds
- Autonomous UGV navigation & kinematics
- Adaptive and variable-resolution mapping

References

- KITTI Vision Benchmark Suite (Velodyne HDL-64E)
- ETH Zurich Robot-Centric Elevation Mapping (P. Fankhauser)
- CMU Robotics Institute Pure Pursuit Kinematics (R. C. Coulter)
- Open3D & PyTorch Deep Learning Documentation
- Research literature on LiDAR semantic segmentation & autonomous foveated navigation

LiDAR 2.5 RESEARCH PIPELINE

LiDAR 2.5 RESEARCH PIPELINE



Verified Project Deliverables & Repository:

- Working 60 FPS HTML5 / Flask Localhost Simulator with moving-car foveation
- Automated Unit Test Suite (5/5 tests passing in 4.0s verifying all modules)
- Open Source GitHub Repository: github.com/AbhayVerma628/SIH26053_Lidar
- Complete 5-panel Matplotlib dashboard, 10 Hz vehicle telemetry log CSV & benchmarks